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Date: May 6, 2026

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Analysis of Large Language Models for Suicide Prevention

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Abstract—In this report, we aim to comprehensively analyse the potential use of LLMs to address global mental health problems, especially suicide prevention, detection, and treatment. We first review the historical technical transition of NLP domain from conventional machine learning methods to the latest Transformer architecture based approaches, while showcasing some challenges to overcome when introducing LLMs into clinical settings. In the benchmark section, Mental RoBERTa demonstrated high accuracy in multiple datasets (0.806 on Dreddit and 0.919 on Mental Health Balanced), showing the effectiveness of domain-specified LLMs. In methodology section, we propose SETE framework utilising LSE to enhance the classification capability and robustness of LLMs towards unseen data.

Index Terms—Large Language Models (LLMs), Natural Language Processing, Zero-shot Learning, Mental Health, Suicide Prevention, Sentiment Classification

1 INTRODUCTION

MENTAL health is one of the critical issues in global health and is a basic human right as it plays an integral role in maintaining our well-being [1]. In 2021, 1.1 billion people were experiencing a mental disorder, including anxiety disorders, depression, bipolar disorder, and etc., which accounted for approximately 15% of all the population around the world. Although there are effective prevention and treatment options, many people are hesitant to have access to such care due to concerns over discrimination, stigma, and violations of human rights [2]. According to The Royal College of Psychiatrists, 43% of adults with mental illness experienced the long waits for treatment, moreover, 23% had to wait more than 12 weeks until they started receiving a treatment. Some people mentioned the exacerbation of their mental health during the wait [3]. This indicates the status quo of the clinical environment where the demands for the treatment exceed the supplies. Especially among mental health issues, prevention of suicide has been a highly prioritized global issue recently. In 2021, 727,000 people were estimated to have lost their lives to suicide, which is equivalent to a global age-standardized suicide rate of 8.9 per 100,000. In addition, it is considered that there are more than 20 suicide attempts for every death by suicide [4]. Despite the fact that there is a good chance to reduce the suicide rate, it has been remaining unchanged due to several factors, such as aforementioned limited service capacity [5]. In this evolving context, large language models (LLMs) draw attention as a potential solution for this urgent

situation, which is capable of facilitating the effective delivery of mental health care [6].

LLMs are able to understand natural language accurately, which is underpinned by the intrinsic artificial neural networks trained on extensive datasets [6]. Due to recent technological advancements in Natural Language Processing (NLP), LLMs have learned to understand the sentences written by humans and their nuance more and more accurately. After the release of Chat Generative Pre-trained Transformer (ChatGPT) in November 2022, ChatGPT has demonstrated quality performance in medical reasoning and diagnosis in clinical domains [7]. Therefore, the psychiatric field has been considered to be a promising area for the application of LLMs, as psychiatric diagnosis relies heavily on verbal narratives from patients that are vague and elusive [7]. However, the deployment of LLMs in mental health also poses several risks, including inconsistencies and hallucinations in the outputs, which might mislead or harm users [8] and, at the same time, raise serious ethical concerns [6]. To address this, the World Health Organization has published a guidance on large multi-modal models, in which they highlighted the importance of data privacy, human oversight, and the principle that AI tools should not replace human professionals [9]. These latent problems that LLMs have in mental health care shed light on the need for a comprehensive and responsible use of LLMs in clinical settings [6]. This report aims to provide a comprehensive analysis of the mechanism of LLMs and their use in the context of clinical settings, especially for the prevention, detection, and treatment of suicide.

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2 LITERATURE REVIEW

2.1 Transition and taxonomy of NLP in mental health

2.1.1 From Traditional Machine Learning to Deep Learning

The flow of NLP evolution started from dictionary/statistics-based methods that pay attention to specific words. After that, traditional machine learning methods that learn data structure emerged before deep learning methods that are capable of maintaining complex contexts and long-term correlations in sentences.

Dictionary/statistics-based methods:

In the early stage, the methods for NLP were primarily based on the number or importance of words used.

LIWC (Linguistic Inquiry and Word Count):

LIWC is a text analysis program developed in 2001, based on the theory that words represent psychological, social, and sentimental states of people. This method calculates the words in the text in a linguistic dimension of approximately 80 to output the word distribution based on categories, such as positive/negative feelings and cognitive processes [10] [11].

TF-IDF (Term Frequency-Inverse Document Frequency): TF-IDF is a traditional method used in the field of information search. This method estimates the importance of words based on the word frequencies and their rarities [12].

Limitation: In these methods, where the word order and the context are ignored, lacks the capability of comprehending the semantics and syntax of the sentence [13].

Traditional Machine Learning Methods: These methods classify or predict the data based on the statistical features.

Regression: Linear regression and Lasso is used to predict values such as sentiment score and psychological metrics based on the features extracted from texts [14] [15].

SVM (Support Vector Machine): SVM is one of the supervised algorithm models that constructs a hyper plane to separate classes in a high dimensional space [15]. In the context of NLP, this method is widely used in word sense disambiguation, information filtering and text automatic classification [16].

Random Forest (RF): Random Forest is one of the ensemble machine learning models that builds multiple decision trees to predict classes or values by selecting the mode class or averaging the continuous values. This method is relatively robust, demonstrating a high accuracy by reducing the overfitting [17] [18].

Limitation: Feature Engineering is necessary to properly function these models, consuming

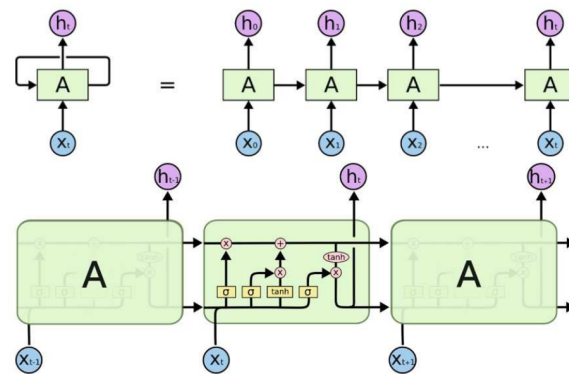


Fig. 1. The architecture of RNN and LSTM: x_t : input data at time step of t , h_t : hidden state (output) at time step of t [21]

enormous resources and time [13].

Deep Learning Methods: Deep learning models capable of direct handling of the data sequence and context have led to a quantum leap in NLP.

RNN (Recurrent Neural Network) 1: RNN is designed to analyse the sequential data such as natural language. Each neuron has a recurrent structure to receive the output from the precedent steps, which enables to encode the sequential information, allowing to recognise the latent semantic relationships between words and the syntax of sentences [13].

LSTM (Long Short-Term Memory) 1: Conventional RNN models suffer from the vanishing gradient problem, limiting its use case where long sentences are handled. LSTM is an extended version of RNN that is capable of maintaining long-term dependencies by introducing a mechanism called gated recurrent unit (GRU) [19] [20].

Advantage: Deep learning models can capture the latent patterns within data by constructing an end-to-end pipeline, directly mapping the outputs from inputs, reducing the amount of time and resources consumed by manual feature engineering [13] [20].

2.1.2 The Emergence of Encoder-based/ Decoder-based and Encoder-Decoder based Models

Recent LLMs are primarily divided into three types based on the focus on the Transformer architecture.

Encoder-based models: Encoder-based models utilise the encoder architecture of Transformer and specialise in learning the semantic representation for each token in the text, enabling to understand the context of whole the sentence [22].

BERT: Masked Language Model (MLM), where the models need to predict masked words based on the context, and Next Sentence Prediction (NSP) was introduced to achieve the state-of-the-art results on sentence-level and token-level natural language processing tasks at the moment [22].

RoBERTa: In RoBERTa paper, authors indicated that BERT model was undertrained. Post-BERT model was trained using more amount of data, bigger batch size, and dynamically changing masking patterns to improve its performance significantly [23].

MentalBERT: Mental BERT is a domain-specific model, which is trained on the mental health dataset collected from various social media, such as Reddit [24].

These models are suitable for the tasks that require understanding the sentences precisely, such as sentence classification, named entity recognition, and question answering tasks. However, these models are not suitable for sentence generating tasks [22].

Decoder-based models: Decoder-based model utilises the decoder architecture of transformer, which predicts the next tokens auto-regressively based on the context to generate sentences. This model uses a causal self-attention mechanism, which prevents the model from attending to the following tokens [22] [25] [26].

GPT series: Generative pre-training was introduced using a wide variety of unlabeled dataset, enabling the model to achieve a general task-agnostic models, outnumbered the conventional task-specific models [27] [25].

LLaMA: LLaMA is a collection of foundation language models trained on publicly available dataset, which aims to achieve the equivalent performance to GPT-3 with fewer number of parameters [28].

These models are suitable for generating tasks, such as creating narratives and answering questions with comprehensive and informative outputs [25].

Encoder-Decoder based models: Encoder-Decoder based models are the standard architecture of transformer, consisting of encoder, which enables the model to understand the input sentences, and decoder, which generates the output according to the input. These models have strength in Sequence-to-Sequence tasks [29] [26].

T5 (Text-to-Text Transfer Transformer): T5 allowed the model to perform various tasks, such as classification, translation, and summarisation, in unified "text-to-text" format, where you only need to input the text to make the model perform the task, before obtaining the text output [26].

BART: BART introduced a denoising autoencoder that corrupts text and learn the model to reconstruct the original sentence. This model demonstrated high performance in text generation tasks, while maintaining the matching capability to RoBERTa in sentence comprehension [29].

2.1.3 The Evolution of Generative and General-Purpose LLMs

While conventional Encoder models, such as BERT, were specified for accurate classification tasks utilising the bidirectional representation [22], Decoder models, such as GPT series, enabled the models to generate sentences auto-regressively based on the context [25]. In terms of the latest LLMs, they are capable of handling more complex tasks, such as reasoning [30]. And also, it has been challenging to collect data related to mental health due to the aspect of privacy and ethical issues, however, LLMs obtained generic knowledge through pre-training, which enables the model to adapt to domain-specific tasks under few-shot or zero-shot condition, demonstrating its potential [28] [27]. The versatility of LLMs improved, however, its hallucination could lead to serious risks for patients [25]. Therefore, there has been a trade-off between the versatility of LLMs and medical accuracy and safety, which means it is necessary to develop more robust and safe LLMs.

2.2 Theoretical Background and Architectures

In this section, we introduce relevant frameworks and architectures that underpin the recent natural language processing.

2.2.1 Foundational Frameworks

End-to-End learning by deep learning: While conventional machine learning, such as SVM and RF, focuses on extracting features from text through manual engineering by humans, deep learning method provided an end-to-end mechanism that enable the models to extract not only such features but also hidden patterns and complex structure in the text automatically [13].

From RNNs to Transformer: In early stage, RNNs were used to handle sequential data, such as natural language. RNNs employ the recurrence link that allows the model to understand the syntax of the sentence and their semantic interaction, while maintaining historical information. However, RNNs experience the vanishing of gradient problem when dealing with long text. To address this issue, GRU and LSTM were introduced [13] [20].

2.2.2 Innovative Architectures

The recent giant leap in LLMs stems from Transformer architecture proposed in 2017. This architecture was innovative in that it removed the RNNs link and CNN and utilised attention mechanism alone [31].

Self-Attention: Transformer is underpinned by the attention mechanism that calculates how much the tokens in different positions in one sentence are related to one another, through which the model can understand the semantics of the sentence. An attention function is represented as below:

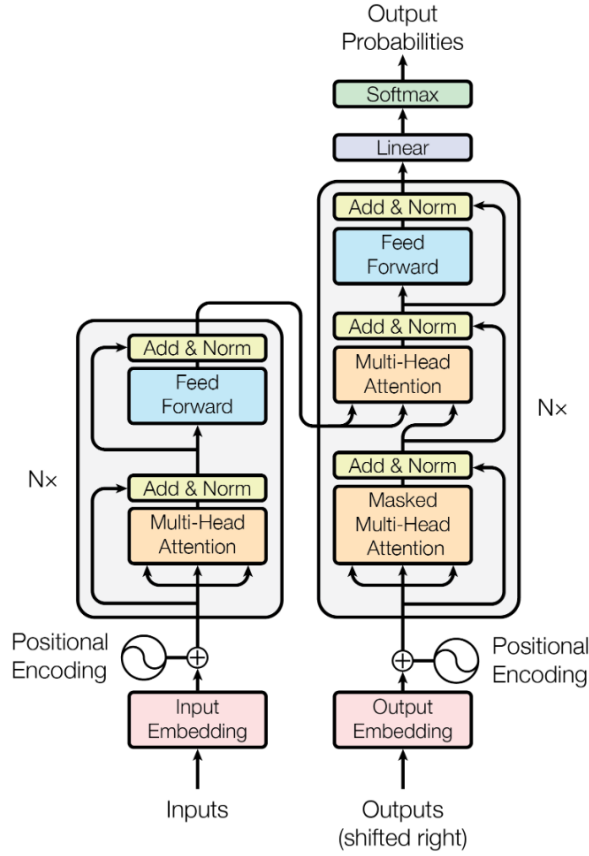


Fig. 2. Transformer architecture [31]

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where Q is a query and K and V are a set of key-value pairs to an output. Output is computed as a weighted sum of value based on query and key. d_k serves as a scaling factor that mitigates gradient vanishing [31].

Multi-Head Attention: Multi-Head Attention is literally the multiple use of self-attention, enabling the model to understand the input sentences from various kind of aspects, such as grammar and what the pronoun refers to. Multi-Head attention is calculated as follows in a similar way to self-attention.

Positional Encoding: Since Transformer do not use recurrence networks, positional encoding was incorporated to ensure the model to hold the sequential information of words. Normally, sine and cosine functions are used to inject the positional information to the embedded vectors.

The Architectures of major pre-trained model: Recent LLMs are primarily divided into three types based on the focus on the Transformer architecture.

Encoder-based (BERT): This model employs multilayer bidirectional Transformer encoder, making it for the model to understand the sentence

from both forward and backward as seen in Fig 4. In the pre-training phase, the model is asked to predict the masked words in a sentence, MLM, or to confirm the consistency of the input sentences, NSP, through which the model can understand the sentence, becoming suitable for tasks, such as sentence classification, named entity recognition, and Q and A [22].

Decoder-based (GPT): This model is based on the decoder architecture of the standard Transformer architecture. The model is trained by predicting the next word in a generated sentence under the condition when given the previous words at the moment (autoregression). This training allows the model to learn our language and its representations, making it suitable for specific downstream task after fine-tuning [25].

Encoder-Decoder based (T5/BART): This model utilises the standard Transformer architecture which employs encoder for understanding the input sentence and decoder for generating sentences according to its semantics. This model integrated various types of NLP tasks into one useful format, text-to-text [26]. Underpinned by this architecture, this model has a good affinity with Sequence-to-Sequence (Seq2Seq) task, such as translation [29].

2.3 Trade-offs and Challenges

Even though the development of LLMs has been remarkable, its application to clinical settings, especially mental health field, poses risks in technological and practical way. In this section, we will discuss the trade-offs and challenges that recent LLMs should handle.

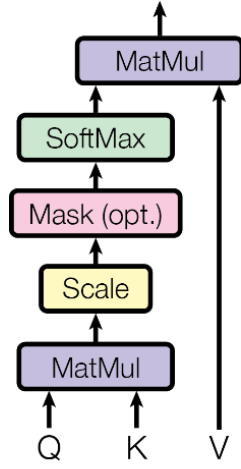
2.3.1 Analysis of technological trade-off

Accuracy and Computational cost: The number of parameters of LLMs and training data heavily affect their performance, demonstrating superior performance in inference, however, it is necessary to have enough resource to achieve it, and that limitates its application to such fields which requires better real-time performance [28].

Interpretability and Performance: Deep learning models, represented by GPT-3, shows high performance, but the process of the models output remain a blackbox, making it difficult to understand and explain [25]. On the other hand, ReHAN (recurrent hierarchical attention net-work), utilising attention mechanism, might provide decent interpretability to cases, but it may experience the trade-off in terms of performance, compared to more complex architectures [19].

Structured dataset and Free-style text: The models, trained on structured dataset tends to perform better than those on no-structured dataset,

Scaled Dot-Product Attention



Multi-Head Attention

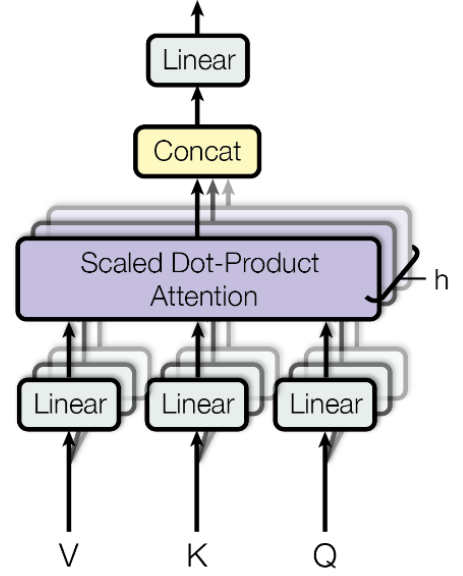


Fig. 3. Scaled Dot-Product Attention represented by equation 1 and Multi-Head Attention [31]

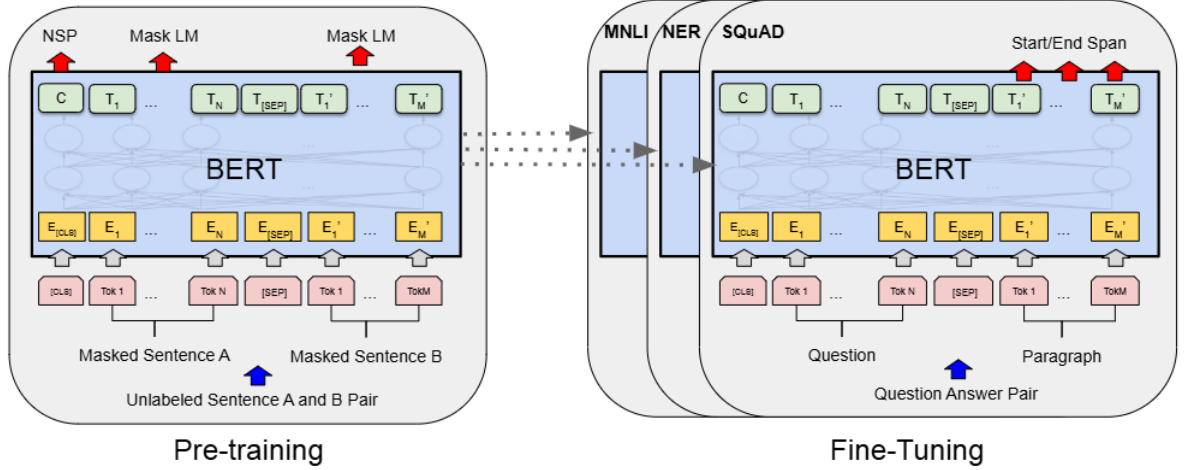


Fig. 4. The architecture of BERT: After pre-training, model is fine-tuned to specific tasks [22].

requiring more manual work from patients and practitioners [32].

2.3.2 Challenges

Prevention of hallucination: Even the latest version of LLMs can generate plausible-looking lie, called hallucination, posing risks for medical advice and assistive diagnosis [28].

Privacy Issues: In the domain of mental health, it is hard to procure a large dataset, considering privacy issues [24].

3 BENCHMARKING RESULTS

In this section, we compare the performance of each model on different datasets. Through this compari-

son, we aim to find the suitable models for methodology part, where we discuss how to approach unknown dataset.

3.1 Dataset

For the benchmarking, we used these three datasets available on the internet.

Dreaddit: This dataset is collected from subreddit in Reddit, one of social media, comprising of 190K posts and made for stress classification. Classification categories include 0 for stress negative, and 1 for stress positive, meaning this is a binary classification task.

Mental Health Balanced: This dataset is also collected from Reddit and made for classification

TABLE 1
Benchmark Summary of Mental Health Classification Models

Dataset	Model	Acc.	F1-M	Prec.	Rec.	CO2 (kg)
Dreaddit	mental/mental-roberta-base	0.806	0.806	0.806	0.806	0.001068
	facebook/bart-base	0.764	0.764	0.765	0.765	0.001162
	mental/mental-bert-base-uncased	0.764	0.763	0.766	0.763	0.001036
	gpt2	0.722	0.721	0.729	0.724	0.001201
	bert-base-uncased	0.694	0.694	0.694	0.694	0.001031
Mental Health Balanced	mental/mental-roberta-base	0.919	0.919	0.920	0.919	0.017689
	mental/mental-bert-base-uncased	0.915	0.915	0.916	0.915	0.017791
	bert-base-uncased	0.909	0.909	0.910	0.909	0.017508
	facebook/bart-base	0.898	0.899	0.902	0.898	0.021568
	gpt2	0.852	0.852	0.855	0.852	0.023874
Reddit 500	mental/mental-bert-base-uncased	0.480	0.335	0.299	0.383	0.001305
	facebook/bart-base	0.380	0.240	0.218	0.279	0.001631
	bert-base-uncased	0.400	0.206	0.160	0.299	0.001302
	mental/mental-roberta-base	0.360	0.185	0.147	0.282	0.001379
	gpt2	0.320	0.098	0.067	0.188	0.001686

tasks. It consists of 13K rows of text data with 13 categories, including "PTSD", "Stress", "Loneliness", "ADHD", and "Anxiety".

Reddit 500: This dataset includes 500 anonymised posts from Reddit. Classification category comprises "Ideation", "Supportive", "Indicator", "Behavior", and "Attempt".

3.2 Evaluation Metrics

Accuracy (ACC.): Accuracy means the ratio of correct answers over the whole prediction, represented by the equation below:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

F1-Macro (F1-M): F1-Macro is calculated by the equation below, aiming to evaluate the model considering the bias in the dataset.

$$F1_i = 2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$$

$$\text{F1-Macro} = \frac{1}{N} \sum_{i=1}^N F1_i$$

Precision and Recall: Precision and recall are calculated by the equations below, respectively. While precision is defined as the ratio of correct prediction over the data predicted as positive, recall is defined as the ratio of true predictions over the actual positive cases.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

CO₂ Emission (kg): This means the amount of carbon dioxide estimated to emit while the model is under training and inference. This metric is introduced due to the SGDs perspective.

3.3 Evaluation Candidates

Following the literature review, we selected 5 potential candidates to execute benchmarking, bert-base-uncased, mental-bert-base-uncased, and mental-roberta-base, from encoder-based model, gpt-2, from decode-based model, bart-base, from encoder-decoder-based model. Comparing these models will provide us with the insight over the differences between foundation architectures.

3.4 Quantitative Analysis

In the Dreaddit dataset, mental-roberta-base model recorded the highest accuracy among these models with 0.806, and also recorded the equivalent values in every evaluation metric. bart-base model and mental-bert-base model posted almost the same value in each metric, leading them to take the second and third place. gpt2 showed 0.722 accuracy with fourth place and only bert-base model lowered 0.700 in accuracy. In the Mental Health Balanced dataset, mental-roberta-base model demonstrated 0.919 in

accuracy, listed in the first place again, also showing the equivalent values in respective metrics. Second place went to mental-bert-base model with accuracy of 0.915 and equivalent values in each metric. Bert-based model took the third place. In this dataset, models with bert architecture demonstrated the high performance compared to other models, indicating that bert architecture has its strength in understanding the sentence. Finally, in the Reddit 500 dataset, the accuracy remained lower than that of other dataset. Even though mental-bert-base model showed 0.480 accuracy and bart-base model showed 0.380, the other metrics got lower than accuracy due to the class imbalance in the dataset. When it comes to CO_2 emission, we can see the difference due to the dataset size, but cannot see a significant difference between models.

3.5 Qualitative Analysis

TABLE 2
Example of sentence and classification by each model

Text:	"Finally let my wife know I was suicidal and she told me to essentially man up and figure it out myself Yeah Im done. She told me to go get help and figure it out myself. No real compassion at all from her at all. I really hate being a man. I know what I must do now.",
true label	Depression
bert-base-uncased	Suicidal
mental-bert-base-uncased	Suicidal
mental-roberta-base	Suicidal
gpt2	Suicidal
bart-base	Suicidal

We extracted 5 cases and the prediction to the case by each model. Table 2 is one of them. In terms of this prediction, all models predict that the sentence is categorised as "Suicidal", even though the true label is "Depression". When we look into the text itself, it contains words, such as suicidal. In addition, the last statement, "I really hate being a man. I know what I must do now", sounds like he is going to commit a suicide, making it more difficult for not only models but also humans to predict the correct label. Considering those facts, all models seemed to perform well. In terms of reddit 500 dataset case, fourth case, we did not list the case in table since the text is too long, however, three models based on bert architecture predicted the correct label, while other two did it wrong. This indicate that the models based on such architecture is more suitable for understanding long sentences and its context. For further details, please refer to the supplementary document.

4 METHODOLOGY

In this section, we first look at the classification result of 10 provided sentiment sentences. After that, we propose Semantically Extended Textual Entailment (SETE) framework that utilising Label Semantic Extension (LSE) based on the reference paper [33] to improve the classification accuracy for unseen mental health dataset.

4.1 Classification Result

For this task, we chose the most potential model, Mental RoBERTa and Mental BERT, based on the result of Benchmark section. When executing this task, the training weight for Reddit 500 was used since their category was similar enough to evaluate this classification task. The result is displayed as seen Table 3.

TABLE 3
Classification result for given cases

Text (true label)	Mental BERT		Mental RoBERTa	
	zero-shot	few-shot	zero-shot	few-shot
1 (attempt)	ideation	ideation	indicator	ideation
2 (attempt)	ideation	ideation	supportive	ideation
3 (behavior)	ideation	ideation	supportive	ideation
4 (behavior)	ideation	ideation	supportive	ideation
5 (ideation)	ideation	ideation	supportive	ideation
6 (ideation)	ideation	ideation	supportive	ideation
7 (indicator)	ideation	ideation	supportive	ideation
8 (indicator)	ideation	ideation	supportive	ideation
9 (safe [supprtive])	indicator	ideation	supportive	ideation
10 (safe [supprtive])	indicator	ideation	supportive	ideation

As we can see in the table, both the models seemed to have problem when predicting unseen and very few data like this case. To address this challenging task, we are proposing the following method.

4.2 Problem Formulation

In the paper [33], the author formulates the zero-shot text classification task as one of the inference problems by natural language inference (NLI). They regarded input text x as premise and category y as hypothesis, turning the task into a textual entailment problem.

4.3 Label Semantic Extension (LSE)

Based on the SETE framework, the label semantics is expanded by the three knowledge base below: Expert knowledge: A couple of core words of the category is collected from reliable sources . ConceptNet knowledge: Using ConceptNet, the model expands its label semantics by extracting 10 related words to the expert knowledge. Contextual knowledge: Using masked language model of BERT, the

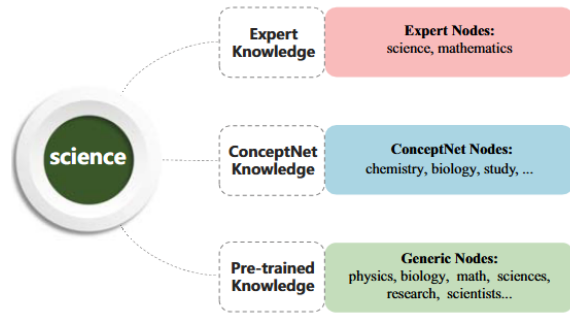


Fig. 5. An illustration of label semantic extension [33].

model generates 50 words that tends to appear in the context of the category. By doing this, the model can understand the semantics of the label from various aspects rather than only a category name.

4.4 Semantically Extended Textual Entailment (SETE)

In the paper, MNLI-BERT was used as an inference model. MNLI-BERT is a transformer base encoder pre-trained on MNLI dataset. The input sequence is given by [CLS] Text [SEP] Extended Label Vocabulary [SEP] and the model calculates the matching score p represented below to predict the label.

$$p = \sigma(W^T m + b)$$

where m is the hidden vector corresponding to [CLS] token in the final layer as the aggregate representation and W and b are the parameters of the linear layer. The prediction is made by picking up the category that shows the highest matching score p .

4.5 Expected Impact on Classification of Given Dataset

By introducing this methodology, the classification capability is expected to get better due to these expectations:

1. High capability towards the entailment problem underpinned by training on MNLI dataset will adapt well to the mental health domain and its slight difference.
2. Enhancing the label semantics will enable the model to handle unseen dataset, leading to the higher robustness to such dataset. This will also prevent false predictions stemming from ignoring the context.

5 AI ETHICS

Australia's AI Ethics Principles serve as a clear guideline for Australian government and companies to develop their AI systems in a responsible

manner. When developing AI systems, they need to abide by the principles and make sure to provide safe, fair, and responsible products for Australian citizens, and to prevent the risk expected by their systems. This also works as a signal to consider ethical aspects of the AI applications. Australia's AI Ethics Principles consist of these 7 items, Human, societal and environmental wellbeing, Human-centred values, Fairness, Privacy protection and security, Reliability and safety, Transparency and explainability, Contestability and Accountability. In this section, we are going to discuss how our proposed system adheres to the principles.

5.1 Human, societal and environmental wellbeing

Our system aims to classify texts related to mental problems, enabling doctors to identify the person who is suffering from such issues. Finding those people as early as possible allows doctors to provide them with proper clinical care, leading to the mentally healthy society. Thus, our system is expected to benefit our society.

5.2 Human-centred values

Our system is designed to help doctors diagnose or find patients with mental health issues, rather than replace them. Doctors can revise the classification results whenever they feel the result of the system has some problems, or they find some weird points in the classification result.

5.3 Privacy protection and security

When training our model, we make sure to use only open source medical dataset properly and anonymously annotated. In addition, when we make the expert knowledge, we also make sure to confirm there is no identifiable information from the knowledge source. We also implement strict security measures in light of possible cyber attacks.

5.4 Transparency and explainability

When our model makes predictions, it will be designed to display the whole sentences and show the evidence linking to its predictions utilising attention mechanism. Finding the place where the attention architecture attend provides clear understanding of our model's prediction.

5.5 Contestability

Upon the release of this product, we open a customer service window on the internet, where customers report issues of our application and submit their feedback to our end. We make the most of them to refine our product, leading to more comfortable application for customers.

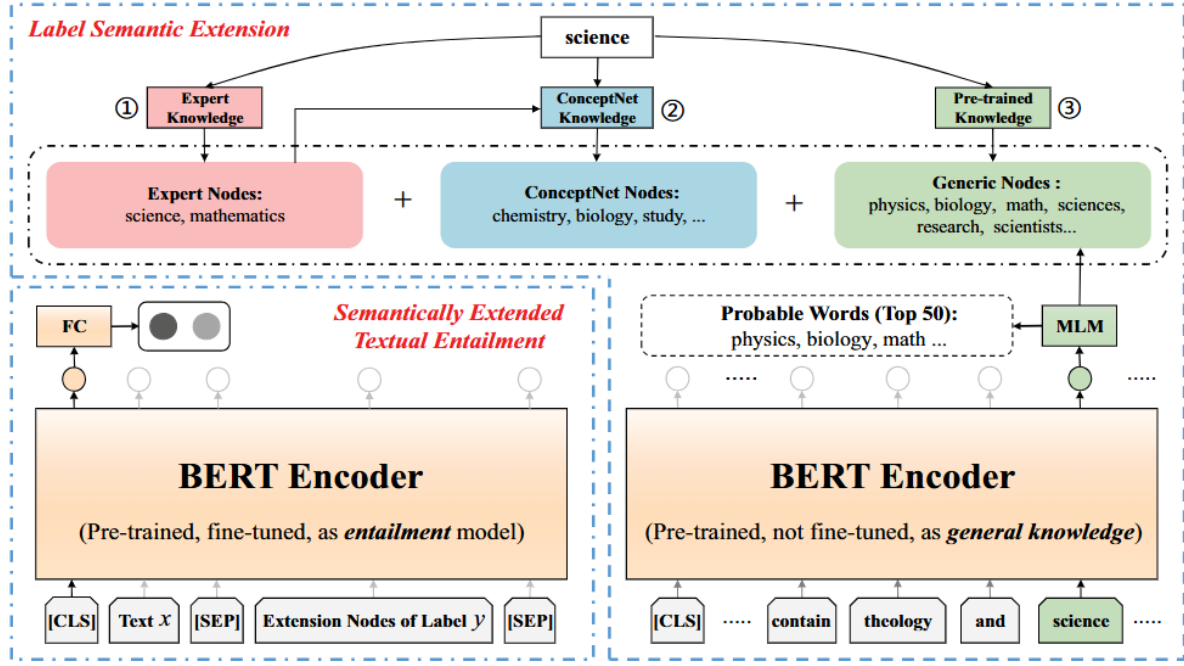


Fig. 6. The framework of zero-shot text classification combining LSE and Textual Entailment Model [33].

5.6 Accountability

We clearly state the whereabouts of the responsibility related to the production of our system, consisting of sales, market, and development section. In addition, we make sure to perform audit periodically to ensure our system adheres to Australia's AI Ethics Principles. In case of malfunction, we immediately analyse the issues and restore the condition to normal state.

6 DISCUSSION

6.1 Sustainable Development Goals (SDGs)

As discussed in the previous section, our product is expected to contribute to the improvement of human wellbeing. The subjects include normal people and those who suffer from mental problems. Based on these expectations, our application will contribute to the 3rd goal of SDGs, Good health and well-being. The estimated carbon emissions of each model are included in Table 1.

6.2 Data Security and Privacy

When processing the sentences, we are recommended to use the LLMs that run on device rather than on websites to protect privacy. In the same sense, it is ideal for our system to be designed using local LLMs, even though there exists a trade-off over performance. We also make sure to use only open source dataset when training our model and creating our own dataset, such as expert knowledge.

6.3 Practical Aspects

When we create this system, it is necessary to prepare the expert knowledge word list, not so many, to enhance the label semantics. However, this makes it easy for our system to adapt new domains not related to medical settings since just picking up some words related to such domain do not require much manual human work.

6.4 Limitations

Our system relies on the capability of LLMs, so it has a limitation for handling new words popping up in dairy lives, since such words did not exist in the past, having prevented our system from learning their relationship with other terms. In terms of computational resources, our system requires more computation as the category increases. This aspect can make it not suitable for large-scale multi-class classification tasks.

6.5 Future Work

In classification task, it is hard to find the clue that results in the final classification. Our system also need to have such capability, such as which augmented word contributed to the final decision most, or how our system thought until it reaches its decision. Implementing this kind of mechanism and enhancing its interpretability will help us engineer expert knowledge most suitable for classification tasks, being expected to lead to better performance.

7 CONCLUSION

In this report, we demonstrated the potential use of LLMs in the context of medical settings through theoretical background and comprehensive benchmark. As a result of evaluation, the pre-trained LLMs specialised in mental health domain excelled in performance compared to general-purpose LLMs, and especially, the series of BERT model has been found to have superior capability in the text classification task. In addition, SETE framework allows LLMs to adapt well to new domains by integrating general knowledge and expert knowledge, showing the potential of LLMs to perform better towards unseen data. While our proposed system is expected to contribute to SDGs' "Good Health and Well-being", intrinsic hallucination risks of LLMs and the demand for such systems run only on local device, due to privacy issues, still remain as challenges. Moreover, enhancing interpretability of these systems is expected to lead to more safe and reliable assistive diagnostic systems.

APPENDIX 1 - VIDEO PRESENTATION

Video presentation is available from the link below:

<https://echo360.net.au/public/media/b4bf5133-4516-421d-95ce-a0ae03b37ea7>

APPENDIX 2 – SUPPLEMENTARY MATERIALS

The source code for benchmarking and other materials are stored in my Github repository and also available from the link below:

https://github.com/Rai-Sanda/comp6011_task2_sub.git

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